

MARKETING ANALYTICS FRAMEWORK, ATTRIBUTION METHODOLOGY & BUSINESS JUSTIFICATION

Multi-Channel Marketing Attribution & ROI Analytics

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1. Executive Summary

1.1 Overview

Modern organizations invest significant resources across multiple marketing channels such as Email Marketing, Paid Advertising, Social Media, Search Engine Optimization (SEO), and Influencer Marketing. While these channels collectively contribute to customer acquisition and revenue generation, determining the actual contribution of each marketing activity remains one of the most challenging aspects of digital marketing.

The objective of this project is to develop a structured Marketing Analytics Framework capable of evaluating campaign performance, measuring marketing effectiveness, and supporting strategic business decision-making through data-driven insights. The solution integrates Python, PostgreSQL, SQL, Microsoft Power BI, and DAX to transform raw marketing data into interactive analytical dashboards that provide meaningful insights into customer acquisition, financial performance, and campaign efficiency.

A Star Schema data warehouse was designed to organize marketing data efficiently, enabling fast analytical querying and scalable dashboard development. Marketing performance was evaluated using campaign-level metrics such as Revenue, Acquisition Cost, Return on Investment (ROI), Click-Through Rate (CTR), Conversion Rate, Cost per Click (CPC), Cost per Acquisition (CPA), Cost per Lead (CPL), and Revenue per Conversion.

The project implements and analytically compares multiple marketing attribution models, including First Touch, Last Touch, Linear, Time Decay, and Position-Based (U-Shaped) Attribution. Since the available dataset contains aggregated campaign-level data rather than customer-level interaction histories, these attribution models are simulated using campaign-level weighting rather than true customer-journey attribution. The interactive Power BI dashboard allows stakeholders to compare campaign performance across all five attribution models, enabling a richer understanding of how different attribution assumptions affect perceived channel and campaign contribution.

The analytical framework provides organizations with actionable insights that support campaign optimization, budget allocation, customer segmentation, profitability analysis, and strategic marketing planning.

1.2 Project Objectives

The primary objectives of this project are:

- Develop a structured marketing analytics framework for evaluating campaign performance.
- Analyze marketing campaigns across multiple acquisition channels.
- Measure financial performance using marketing KPIs.
- Design and justify an appropriate marketing attribution methodology.
- Identify high-performing and underperforming campaigns.
- Support strategic budget allocation using quantitative analysis.
- Generate actionable business recommendations through interactive dashboards.
- Demonstrate the practical application of Business Intelligence technologies in marketing analytics.

1.3 Business Scope

The analytical framework focuses on evaluating marketing campaigns across multiple business dimensions, including:

- Brand Performance
- Marketing Channel Effectiveness
- Customer Segmentation
- Campaign Type Performance
- Financial Profitability
- Marketing Return on Investment
- Customer Acquisition Efficiency
- Strategic Marketing Recommendations

The framework is designed to assist business stakeholders in making informed marketing decisions based on measurable performance indicators rather than assumptions.

1.4 Expected Business Outcomes

The implementation of this project enables organizations to:

- Evaluate campaign effectiveness across multiple acquisition channels.
- Improve marketing return on investment through performance-based optimization.
- Identify profitable customer segments and marketing channels.
- Reduce customer acquisition costs by optimizing campaign strategies.
- Generate executive-level dashboards for business reporting.
- Support evidence-based marketing decisions using Business Intelligence techniques.

Figure 1. Overall Marketing Analytics Framework

1. Raw Marketing Dataset
2. Python Data Cleaning & ETL
3. PostgreSQL Data Warehouse (Star Schema)
4. SQL Data Analysis
5. Power BI Data Modelling
6. DAX KPI Calculations
7. Interactive Business Dashboard
8. Business Insights & Recommendations

Figure 1: End-to-end flow from raw data to business insights

1.5 Document Structure

This document is organized into the following sections:

- Executive Summary
- Project Overview
- Analytical Approach
- Data Model
- Attribution Methodology
- Business Assumptions
- KPI Framework
- Attribution Model Comparison
- Business Insights
- Optimization Strategy
- Limitations
- Future Scope
- Conclusion

Each section provides a detailed explanation of the analytical framework, technical implementation, and business rationale adopted throughout the project.

2. Project Overview & Analytical Approach

2.1 Project Overview

Marketing organizations rely on multiple acquisition channels to attract, engage, and convert potential customers. These channels include Email Marketing, Paid Advertising, Social Media Marketing, Search Engine Optimization (SEO), and Influencer Marketing. While these campaigns generate large volumes of marketing data, organizations often struggle to identify which channels contribute most effectively to customer conversions and revenue generation.

To address this challenge, this project develops a comprehensive Marketing Analytics Framework that integrates data engineering, data warehousing, business intelligence, and marketing analytics into a single analytical solution. The framework transforms raw marketing campaign data into meaningful business insights through a structured pipeline consisting of Python-based data preprocessing, PostgreSQL data warehousing, SQL-based analytical querying, and Microsoft Power BI dashboard visualization.

The implemented solution enables marketing managers and business stakeholders to evaluate campaign effectiveness, compare acquisition channels, measure financial performance, and optimize future marketing strategies using measurable business indicators rather than intuition.

2.2 Business Problem

Organizations invest substantial marketing budgets across multiple channels with the expectation of maximizing customer acquisition and revenue growth. However, several challenges limit their ability to evaluate campaign performance effectively:

- Difficulty in identifying the most profitable marketing channels.
- Lack of visibility into campaign efficiency and customer acquisition costs.
- Challenges in allocating marketing budgets based on measurable performance.
- Limited understanding of customer segment behavior across campaigns.
- Absence of a structured attribution methodology to measure campaign contribution.
- Inefficient reporting caused by fragmented marketing data.

Without a centralized analytics framework, marketing decisions are frequently based on assumptions rather than quantitative evidence, leading to inefficient resource allocation and reduced return on investment.

2.3 Solution Architecture

To overcome these challenges, an end-to-end Business Intelligence solution was developed using modern data analytics technologies. The solution follows a structured workflow that converts raw marketing data into executive-level dashboards and actionable business recommendations.

The architecture consists of six major stages:

- Data Preparation using Python.
- Data Storage and Modeling using PostgreSQL.
- Business Analysis using SQL.
- Dashboard Development using Microsoft Power BI.
- Advanced Attribution Modeling across First Touch, Last Touch, Linear, Time Decay, and Position-Based methodologies.
- Business KPI Framework consolidating financial and attribution-derived metrics.

Each stage contributes to building a scalable and efficient marketing analytics framework capable of supporting strategic business decisions.

Figure 2. End-to-End Analytical Workflow

1. Raw Marketing Dataset
2. Python Data Cleaning & Feature Engineering
3. PostgreSQL Data Warehouse
4. SQL Analytics
5. Power BI Dashboard
6. Advanced Attribution Modeling
7. Business KPI Framework
8. Strategic Business Insights

Figure 2: Detailed analytical workflow from data collection to insight generation

2.4 Analytical Workflow

The project follows a structured analytical workflow to ensure data accuracy, consistency, and meaningful business interpretation.

Step 1 – Data Collection

A structured marketing campaign dataset containing campaign-level performance metrics was used as the foundation for analysis. The dataset includes financial, operational, and customer-related marketing metrics required to evaluate campaign effectiveness.

Step 2 – Data Cleaning and Preprocessing

Python was used to clean and preprocess the dataset before loading it into the analytical database. The preprocessing stage involved handling missing values, validating data types, correcting inconsistencies, removing duplicate records, and preparing the dataset for downstream analysis.

This stage ensured high data quality and improved the reliability of subsequent analyses.

Step 3 – Database Design

The cleaned dataset was imported into PostgreSQL, where it was organized using a Star Schema data warehouse design. Fact and dimension tables were created to support efficient querying and scalable reporting.

The Star Schema improves analytical performance by separating transactional campaign metrics from descriptive business dimensions.

Step 4 – SQL-Based Business Analysis

A comprehensive collection of SQL queries was developed to analyze marketing performance across multiple business dimensions. The analysis covered revenue generation, customer acquisition, campaign profitability, channel effectiveness, brand performance, customer segmentation, and financial trends. Advanced SQL techniques, including aggregation functions, Common Table Expressions (CTEs), window functions, ranking functions, and analytical views, were used to generate meaningful business insights. The analytical workflow includes 37 professional SQL queries organized into thematic sections for campaign, brand, customer, channel, financial, and executive analysis.

Step 5 – Dashboard Development

Microsoft Power BI was used to build an interactive Business Intelligence dashboard. Data modeling, DAX measures, slicers, KPI cards, and interactive visualizations were implemented to provide stakeholders with a comprehensive overview of marketing performance.

The dashboard enables users to analyze campaign effectiveness dynamically across brands, customer segments, channels, campaign types, and time periods.

Step 6 – Advanced Attribution Modeling

Five attribution models — First Touch, Last Touch, Linear, Time Decay, and Position-Based (U-Shaped) — were implemented using campaign-level weighting logic and compared to evaluate how attribution assumptions affect measured campaign and channel contribution.

Step 7 – Business KPI Framework

Financial KPIs and attribution outputs were consolidated into a unified Business KPI Framework, forming the basis for the Power BI dashboard and downstream business insights.

Step 8 – Business Insights

The analytical findings were interpreted to identify:

- High-performing marketing campaigns.
- Profitable customer segments.
- Effective acquisition channels.
- Financial performance trends.
- Budget optimization opportunities.
- Strategic growth opportunities.

These insights formed the basis for the final optimization strategy and executive recommendations.

2.5 Analytical Deliverables

The project delivers the following analytical outputs:

- Cleaned and validated marketing dataset.
- PostgreSQL marketing data warehouse.
- SQL analytical queries.
- Interactive Power BI dashboard.
- Multi-Attribution Modeling Notebook.
- Attribution comparison analysis across five models.
- Financial KPI framework.
- PostgreSQL analytical views.
- PostgreSQL stored functions.
- PostgreSQL triggers.
- Enterprise SQL warehouse.
- Executive Power BI dashboard (5 pages).
- Business insights and strategic recommendations.
- Optimization strategy supported by quantitative analysis.

Together, these deliverables provide a comprehensive decision-support system for evaluating multi-channel marketing performance.

3. Data Model Design

3.1 Introduction

An efficient data model forms the foundation of every successful Business Intelligence solution. Marketing organizations generate large volumes of transactional data across multiple acquisition channels, making it essential to organize information in a structured format that supports fast analytical queries and interactive reporting.

To address this requirement, the project follows a Star Schema as the conceptual analytical model guiding Business Intelligence design. The Star Schema is one of the most widely used dimensional modeling techniques in Business Intelligence because it separates measurable business facts from descriptive business dimensions. This design improves query performance, simplifies dashboard development, and enables scalable analytical reporting. In the implemented PostgreSQL warehouse, this conceptual model is realized through a centralized marketing_campaigns fact table supported by analytical views, SQL functions, and triggers, together with semantic modeling within Power BI, rather than as physically separate dimension tables.

Conceptually, the Star Schema organizes marketing campaign data into a central fact table surrounded by multiple logical dimensions, allowing efficient analysis across brands, channels, campaign types, customer segments, languages, and time. These dimensions are realized in the implemented warehouse as attributes, analytical views, and SQL functions built around the central fact table, rather than as physically separate dimension tables.

3.2 Star Schema Overview

The conceptual Star Schema consists of one central Fact Table that stores quantitative marketing metrics and six surrounding logical Dimensions that provide descriptive business information. In the implemented PostgreSQL warehouse, this conceptual design is realized through a single centralized fact table supported by analytical views, SQL functions, and triggers rather than physically separate dimension tables.

The central fact table contains measurable campaign performance indicators such as revenue, acquisition cost, conversions, clicks, leads, ROI, CTR, CPA, CPC, and CPL.

The dimension tables describe different perspectives through which marketing performance can be analyzed. This structure minimizes data redundancy while improving analytical flexibility.

Figure 3. Star Schema Design

Dimension Table	Connects To
dim_brand	fact_marketing_campaigns
dim_channel	fact_marketing_campaigns
dim_campaign_type	fact_marketing_campaigns
dim_customer_segment	fact_marketing_campaigns
dim_language	fact_marketing_campaigns
dim_date	fact_marketing_campaigns

Figure 3: Six conceptual dimensions surrounding the central fact table (star topology)

3.3 Fact Table

fact_marketing_campaigns

The fact_marketing_campaigns table serves as the central repository for all measurable campaign performance metrics. Each record represents an individual marketing campaign and stores the quantitative values required for performance analysis and business reporting.

The fact table includes the following key metrics:

- Revenue
- Acquisition Cost
- Clicks
- Leads
- Conversions
- Return on Investment (ROI)
- Click-Through Rate (CTR)
- Cost Per Click (CPC)
- Cost Per Acquisition (CPA)
- Cost Per Lead (CPL)
- Conversion Rate
- Revenue per Conversion

These metrics are used throughout the SQL analysis, DAX calculations, and Power BI dashboards to evaluate campaign effectiveness and financial performance.

3.4 Dimension Tables

To support multidimensional analysis, the analytical model defines six conceptual dimensions. Rather than being created as physically separate tables, these dimensions are realized in the PostgreSQL warehouse as attributes of the centralized marketing_campaigns fact table, together with supporting analytical views and SQL functions, and as semantic dimensions within the Power BI data model.

dim_brand

Stores information about the brands included in the analysis, enabling comparison of campaign performance across different business portfolios.

Example attributes:

- Brand ID
- Brand Name

dim_channel

Contains information about customer acquisition channels used by marketing campaigns. Examples include:

- Email Marketing
- Paid Advertising
- Instagram
- Facebook
- YouTube
- SEO

This dimension supports channel-level ROI and performance analysis.

dim_campaign_type

Categorizes campaigns according to marketing strategy. Examples include:

- Paid Ads
- Email Campaign
- Social Media
- Influencer Marketing
- SEO

This dimension enables campaign-type comparison and profitability analysis.

dim_customer_segment

Represents different target audiences served by marketing campaigns. Examples include:

- College Students
- Working Professionals
- Homemakers
- Premium Shoppers
- Youth

Customer segmentation enables organizations to identify the most profitable customer groups.

dim_language

Stores campaign language information. Examples include:

- English
- Hindi
- Tamil
- Bengali

Language-based analysis helps evaluate the effectiveness of localized marketing campaigns.

dim_date

Provides the temporal dimension required for trend analysis. Attributes include:

- Date
- Month
- Quarter
- Year

This dimension supports monthly revenue trends, seasonal analysis, and year-over-year comparisons.

3.5 Relationships

Conceptually, the fact_marketing_campaigns table is linked to each dimension through a one-to-many relationship. In the implemented warehouse, these relationships are expressed through attributes on the centralized fact table together with supporting PostgreSQL views and functions, and through semantic relationships defined within the Power BI data model.

This design allows marketing performance to be analyzed from multiple business perspectives while avoiding unnecessary duplication of descriptive information. The relationship model also improves dashboard performance by reducing query complexity.

Figure 4. Relationship Structure

Dimension	Cardinality	Fact Table
Brand	1 : Many	fact_marketing_campaigns
Channel	1 : Many	fact_marketing_campaigns
Campaign Type	1 : Many	fact_marketing_campaigns
Customer Segment	1 : Many	fact_marketing_campaigns
Language	1 : Many	fact_marketing_campaigns
Date	1 : Many	fact_marketing_campaigns

Figure 4: One-to-many relationships between dimensions and the fact table

3.6 Why Star Schema Was Selected

The Star Schema was selected because it offers several advantages for Business Intelligence and marketing analytics applications. Key benefits include:

- Simplifies analytical queries.
- Improves SQL query performance.
- Optimizes Power BI data modeling.
- Reduces data redundancy.
- Supports scalable dashboard development.
- Enables multidimensional business analysis.
- Simplifies DAX calculations.
- Enhances maintainability of the data warehouse.

These advantages make the Star Schema particularly suitable for marketing analytics projects involving large volumes of campaign performance data.

3.7 Data Model Benefits

The implemented data model provides significant value for business analysis by enabling:

- Brand-wise performance evaluation.
- Channel-wise ROI analysis.
- Customer segment comparison.
- Campaign profitability analysis.
- Financial performance tracking.
- Time-series trend analysis.
- Executive reporting through interactive dashboards.

4. Marketing Attribution Methodology

4.1 Introduction

Marketing attribution is the process of assigning value to marketing channels and campaigns based on their contribution toward customer conversions and revenue generation. Organizations use attribution methodologies to evaluate marketing effectiveness, understand customer acquisition behavior, and optimize budget allocation across different marketing channels.

As digital marketing ecosystems continue to evolve, customers often interact with multiple campaigns before making a purchase. These interactions may include email campaigns, social media advertisements, paid advertisements, influencer promotions, search engine marketing, and other digital touchpoints. Marketing attribution attempts to determine how much each interaction contributes to the final conversion.

The effectiveness of any attribution model depends on the availability of customer interaction data. Organizations with access to complete customer journey information can implement advanced multi-touch attribution models, whereas organizations with campaign-level aggregated data require a simplified but practical attribution methodology.

4.2 Attribution Methodology Implemented

The project implements and analytically compares five distinct marketing attribution methodologies: First Touch, Last Touch, Linear, Time Decay, and Position-Based (U-Shaped) Attribution.

First Touch Attribution

One hundred percent of the credit for a conversion is assigned to the first recorded interaction associated with a campaign.

Last Touch Attribution

One hundred percent of the credit for a conversion is assigned to the final recorded interaction associated with a campaign.

Linear Attribution

Credit for a conversion is distributed equally across all interactions associated with a campaign.

Time Decay Attribution

Progressively greater credit is assigned to interactions occurring closer in time to the conversion, giving more weight to recent marketing activity.

Position-Based (U-Shaped) Attribution

The first and last interactions receive the largest share of credit, while the remaining credit is distributed across the middle interactions.

Because customer-level journeys are unavailable in the dataset, all five models are implemented using campaign-level weighting assumptions rather than true multi-touch, customer-journey attribution. Each model redistributes a campaign's recorded conversions and revenue according to its respective weighting logic, enabling analytical comparison without requiring individual customer interaction histories.

The purpose of implementing and comparing multiple attribution models is to:

- Compare attribution strategies and their effect on measured campaign performance.
- Evaluate channel contribution under different attribution assumptions.
- Demonstrate attribution modeling concepts within a Business Intelligence framework.
- Support business decision-making with a range of attribution perspectives rather than a single fixed view.

Under each model, campaigns are analyzed using measurable business indicators such as:

- Revenue
- Acquisition Cost
- ROI
- CTR
- CPC
- CPA
- CPL
- Conversion Rate
- Revenue per Conversion

This methodology enables direct comparison of campaign effectiveness across attribution strategies while maintaining consistency with the available campaign-level dataset.

4.3 Why Multiple Attribution Models Were Implemented

The decision to implement and compare multiple attribution models — First Touch, Last Touch, Linear, Time Decay, and Position-Based — was based on both technical and business considerations.

The available dataset contains campaign-level aggregated performance metrics but does not include customer-level interaction histories. As a result, individual customer journeys cannot be reconstructed.

Implementing advanced customer-level multi-touch attribution without journey data would require assumptions that could reduce the reliability of the analysis. Instead, campaign-level weighting is used to simulate how each attribution model would redistribute credit, allowing the five methodologies to be compared on a common, transparent basis.

Implementing and comparing multiple attribution models provides several practical advantages:

- Simple implementation.
- Easy interpretation for business users.
- Direct comparison of campaign performance across attribution strategies.
- Reliable financial analysis.
- Suitable for executive reporting.
- Supports campaign optimization decisions.
- Consistent with the available dataset.

Although simplified, the campaign-level weighting approach effectively satisfies the analytical objectives of this project while maintaining methodological transparency across all five attribution models.

4.4 Attribution Workflow

The attribution process follows a structured analytical workflow beginning with campaign execution and ending with strategic business recommendations.

Figure 5. Campaign-Level Attribution Workflow

1. Marketing Campaign
2. Impressions
3. Clicks
4. Leads
5. Conversions
6. Revenue Generated
7. Attribution Logic
8. KPI Calculations
9. Business Intelligence
10. Strategic Business Decisions

Figure 5: Attribution logic applied from campaign impression through to strategic decisions

Each campaign is evaluated independently using campaign performance metrics. The calculated KPIs are visualized within Microsoft Power BI dashboards, enabling stakeholders to compare marketing effectiveness across brands, customer segments, campaign types, acquisition channels, languages, and time periods.

4.5 Business Assumptions

The attribution framework is based on the following analytical assumptions:

- Each conversion belongs to one marketing campaign.
- Revenue is fully attributed to the campaign generating the recorded conversion.
- Acquisition Cost represents Cost Per Acquisition (CPA) rather than total campaign spend. Total Campaign Spend is derived as Acquisition Cost multiplied by Conversions, and this assumption is applied consistently throughout the Python preprocessing, SQL analysis, Power BI dashboards, and financial KPI calculations.
- Campaigns are evaluated independently.
- Customer interaction history is unavailable.
- Cross-device customer tracking is not available.
- Sequential customer journeys cannot be reconstructed.
- Campaign performance metrics accurately represent marketing effectiveness.
- All five attribution models are simulated using campaign-level weighting rather than true customer-level, multi-touch attribution.

These assumptions ensure analytical consistency while remaining aligned with the available dataset.

4.6 Attribution Logic

The implemented attribution logic follows a distinct weighting rule for each of the five models.

Attribution Rule: Under Last Touch, the campaign receives 100% credit for its recorded conversions and revenue. Under First Touch, credit is instead assigned to the campaign representing the first touchpoint. Under Linear, credit is split evenly; under Time Decay, more recent touchpoints receive greater weight; and under Position-Based, the first and last touchpoints receive the largest share, with the remainder distributed across the middle touchpoints. In all cases, weighting is applied at the campaign level rather than the individual customer level.

These rules enable objective comparison of marketing campaigns across different attribution assumptions, without requiring customer-level interaction histories.

The calculated campaign metrics are subsequently used to identify:

- High-performing campaigns.
- Underperforming campaigns.
- Efficient acquisition channels.
- Profitable customer segments.
- Budget optimization opportunities.
- Marketing growth strategies.

4.7 Marketing KPI Framework

The project evaluates campaign performance using multiple Key Performance Indicators (KPIs).

KPI	Formula	Business Interpretation
Revenue	Revenue Generated	Measures campaign income
Campaign Spend	Acquisition Cost $\times$ Conversions	Measures total campaign investment derived from CPA
Net Profit	Revenue $-$ Campaign Spend	Measures campaign profitability
ROI	Net Profit $\div$ Campaign Spend	Measures return on investment
ROAS	Revenue $\div$ Campaign Spend	Measures gross revenue generated per unit of marketing investment
CTR	Clicks $\div$ Impressions $\times$ 100	Measures advertisement engagement
Conversion Rate	Conversions $\div$ Clicks $\times$ 100	Measures customer conversion efficiency
CPC	Campaign Spend $\div$ Clicks	Measures advertising cost per click
CPA	Campaign Spend $\div$ Conversions	Measures customer acquisition cost
CPL	Campaign Spend $\div$ Leads	Measures lead generation cost
Revenue per Conversion	Revenue $\div$ Conversions	Measures average customer value

4.8 Additional KPI Considerations

The project scope also considered several additional marketing KPIs. Their availability depends on the granularity of the underlying business data.

Return on Advertising Spend (ROAS)

ROAS measures gross revenue generated per unit of marketing investment and complements ROI, which measures net profitability. ROAS is implemented in this framework because Campaign Spend is derived as Acquisition Cost (CPA) multiplied by Conversions, giving a usable denominator: $\text{ROAS} = \text{Revenue} \div \text{Campaign Spend}$. Together, ROI and ROAS give stakeholders both a profitability view and a gross-return view of marketing investment.

Customer Acquisition Cost (CAC)

True Customer Acquisition Cost cannot be calculated because customer-level acquisition data — such as individual customer records and per-customer spend — is unavailable in the dataset. Campaign-level CPA is used as the closest available approximation, since every recorded conversion represents a completed customer acquisition at the campaign level.

Customer Lifetime Value (CLV)

Customer Lifetime Value estimates the total revenue generated by a customer throughout the business relationship. The available dataset does not include:

- Customer IDs
- Repeat purchases
- Customer retention
- Subscription history
- Purchase frequency

Therefore, Customer Lifetime Value could not be calculated within the scope of this project.

5. Business Insights, Recommendations & Optimization Strategy

5.1 Introduction

This chapter translates the analytical framework and attribution methodology into concrete business insights, strategic recommendations, and an optimization roadmap for improving overall marketing performance.

5.2 Key Business Insights

1. Revenue Performance

The revenue analysis indicates that marketing performance varies significantly across acquisition channels, campaign types, and customer segments. Campaigns with higher engagement and stronger conversion rates consistently generate greater revenue, demonstrating the importance of optimizing customer targeting and campaign execution.

Revenue trend analysis also highlights seasonal variations, indicating that marketing performance is influenced by customer purchasing behavior during different periods of the year.

2. Campaign Effectiveness

Campaign-level analysis enables the identification of both high-performing and underperforming campaigns. High-performing campaigns demonstrate:

- Strong ROI
- Higher conversion rates
- Lower acquisition costs
- Better customer engagement

Conversely, campaigns with high acquisition costs and lower conversion efficiency represent opportunities for optimization and budget reallocation.

3. Customer Segment Analysis

Customer segmentation reveals that different customer groups respond differently to marketing campaigns. Some customer segments consistently generate higher revenue and better conversion rates, while others require significantly greater acquisition costs to achieve comparable business outcomes.

Understanding customer segment behavior enables organizations to personalize marketing strategies and improve resource allocation.

4. Channel Performance

Marketing channels contribute differently to campaign success. The analytical framework enables comparison of channel performance using Revenue, ROI, CTR, CPC, CPA, and Conversion Rate.

Channels demonstrating high revenue generation together with low acquisition costs represent the most efficient investment opportunities. Channels with consistently poor ROI should be reviewed to determine whether campaign optimization or budget reduction is appropriate.

5. Financial Performance

Financial analysis provides comprehensive visibility into marketing profitability. Key financial indicators include:

- Revenue
- Acquisition Cost
- Net Profit
- ROI
- Revenue per Conversion

Monitoring these indicators enables organizations to balance marketing investment with financial return while maintaining sustainable business growth.

6. Marketing Attribution Insights

The comparison of five attribution models — First Touch, Last Touch, Linear, Time Decay, and Position-Based — reveals meaningful differences in how campaign and channel contribution is measured depending on the attribution assumptions applied. Campaigns that appear strongest under Last Touch may rank differently under Linear or Time Decay weighting, highlighting how attribution methodology choice materially affects perceived channel contribution, budget allocation sensitivity, and campaign prioritization.

Although the campaign-level weighting approach does not reconstruct complete customer journeys, comparing multiple models provides sufficient analytical depth to evaluate campaign profitability from several perspectives and support strategic marketing decisions.

5.3 Strategic Recommendations

Based on the analytical findings, several strategic recommendations are proposed to improve marketing performance.

Optimize Marketing Budget Allocation

Organizations should prioritize investment in campaigns demonstrating consistently high ROI and strong conversion efficiency. Campaigns generating lower returns should be reviewed, optimized, or discontinued to maximize marketing effectiveness.

Strengthen Customer Segmentation

Marketing campaigns should be tailored to the preferences and purchasing behavior of high-value customer segments. Segment-specific messaging and personalized marketing strategies can improve conversion rates while reducing acquisition costs.

Improve Channel Mix

Marketing investments should be distributed across channels according to measurable business performance rather than historical spending patterns. Combining complementary acquisition channels can improve customer reach and increase overall campaign effectiveness.

Focus on Data-Driven Decision Making

Marketing decisions should be guided by analytical evidence rather than intuition. Interactive dashboards and KPI monitoring should become part of routine business reporting to support continuous campaign optimization.

Continuous Performance Monitoring

Marketing performance should be monitored regularly using standardized KPIs. Continuous monitoring enables organizations to identify emerging trends, respond quickly to underperforming campaigns, and capitalize on new growth opportunities.

5.4 Optimization Strategy

The final optimization strategy developed for this project focuses on improving overall marketing efficiency through measurable business actions. The strategy consists of four key pillars:

Pillar	Description
Growth Opportunity	Increase investment in campaigns and acquisition channels demonstrating consistently high ROI and revenue contribution.
Customer Focus	Prioritize marketing efforts toward high-value customer segments that exhibit stronger engagement and higher conversion potential.
Budget Optimization	Reallocate marketing budgets from underperforming campaigns to campaigns delivering superior financial returns and customer acquisition efficiency.
Future Strategy	Expand campaign localization, improve omnichannel marketing integration, automate performance monitoring, and incorporate predictive analytics.

Figure 6. Strategic Decision Framework

1. Marketing Analytics
2. Campaign Evaluation
3. Marketing Attribution
4. Business Insights
5. Strategic Recommendations
6. Budget Optimization
7. Continuous Performance Improvement

Figure 6: From raw analytics to continuous performance improvement

5.5 Future Scope

While the current framework provides a robust foundation for campaign-level marketing analytics and multi-model attribution comparison, several extensions could further enhance its analytical depth and business value:

- Markov Chain Attribution, to model probabilistic customer paths and removal-effect channel contribution.
- Shapley Value Attribution, to fairly distribute conversion credit based on cooperative game theory.
- Customer Journey Analytics, incorporating sequential, timestamped touchpoint data.
- Customer-level Multi-touch Attribution, enabled by access to individual customer interaction histories.
- Real-time Marketing Analytics, supporting live campaign monitoring and rapid optimization.
- Predictive Attribution using Machine Learning, to forecast conversion likelihood and channel contribution.
- Cloud Data Warehouse Integration, for scalable storage and cross-platform analytics.
- Automated ETL Pipelines, to streamline data ingestion, transformation, and refresh cycles.

5.6 Conclusion

This project successfully developed a comprehensive Marketing Analytics Framework capable of evaluating campaign performance, measuring marketing effectiveness, and supporting data-driven decision-making.

By integrating Python, PostgreSQL, SQL, Microsoft Power BI, DAX, and multi-attribution modeling, the framework transforms raw marketing campaign data into meaningful business intelligence that supports campaign optimization, customer segmentation, financial analysis, and strategic planning.

Five attribution models — First Touch, Last Touch, Linear, Time Decay, and Position-Based — were implemented and compared using campaign-level weighting, providing a practical and transparent approach for evaluating campaign performance from multiple attribution perspectives within the constraints of the available dataset. Financial KPIs were corrected using Campaign Spend derived from CPA, enabling accurate ROI and ROAS calculations. Although true customer-level, multi-touch attribution would require customer-level interaction histories, the implemented approach effectively satisfies the analytical objectives of the project while maintaining methodological consistency.

Overall, the project demonstrates how Business Intelligence technologies can enable organizations to optimize marketing investments, improve operational efficiency, and support sustainable business growth through evidence-based decision-making. The interactive dashboards support strategic marketing decisions by making campaign, channel, and attribution-model comparisons directly accessible to business stakeholders, and the completed solution demonstrates modern Business Intelligence, Marketing Analytics, SQL Engineering, and Attribution Analysis practice.